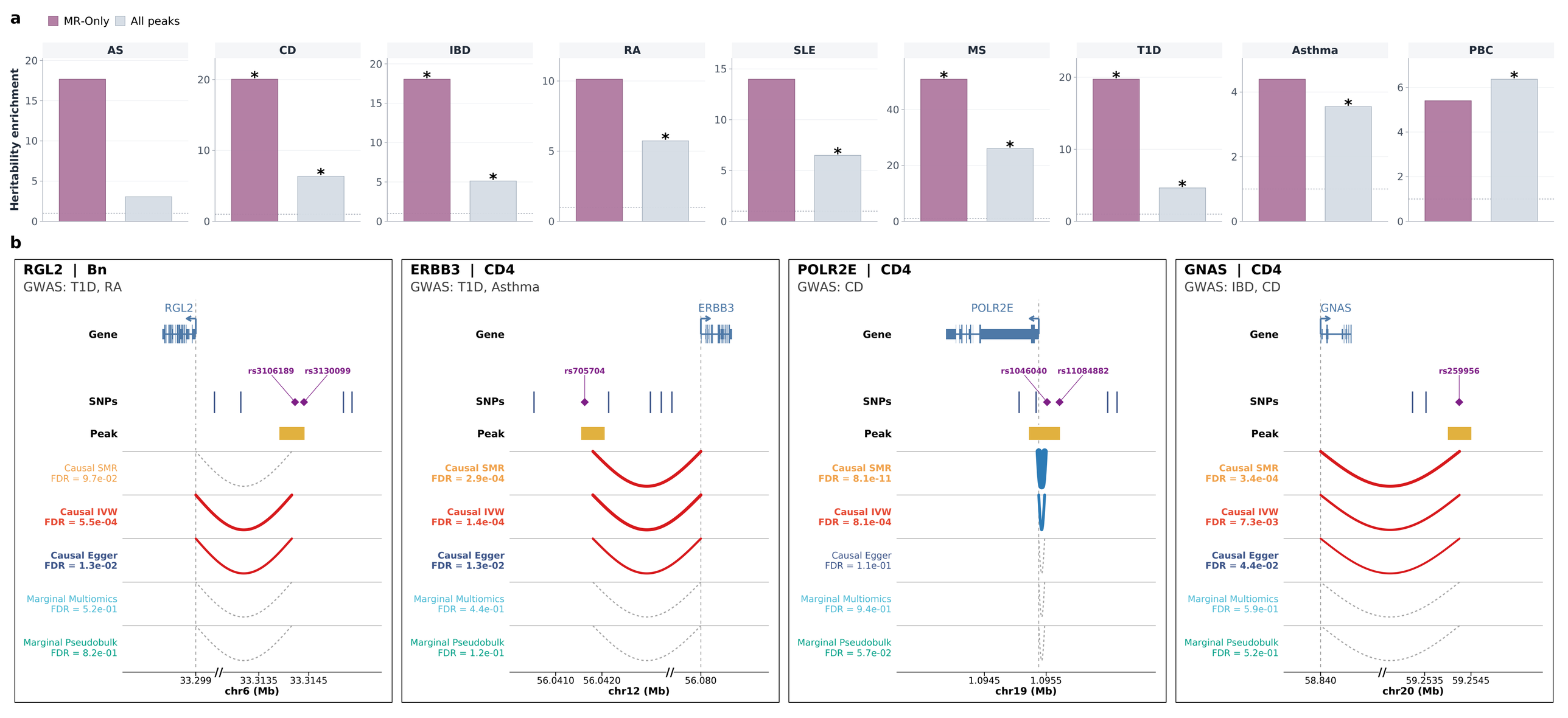
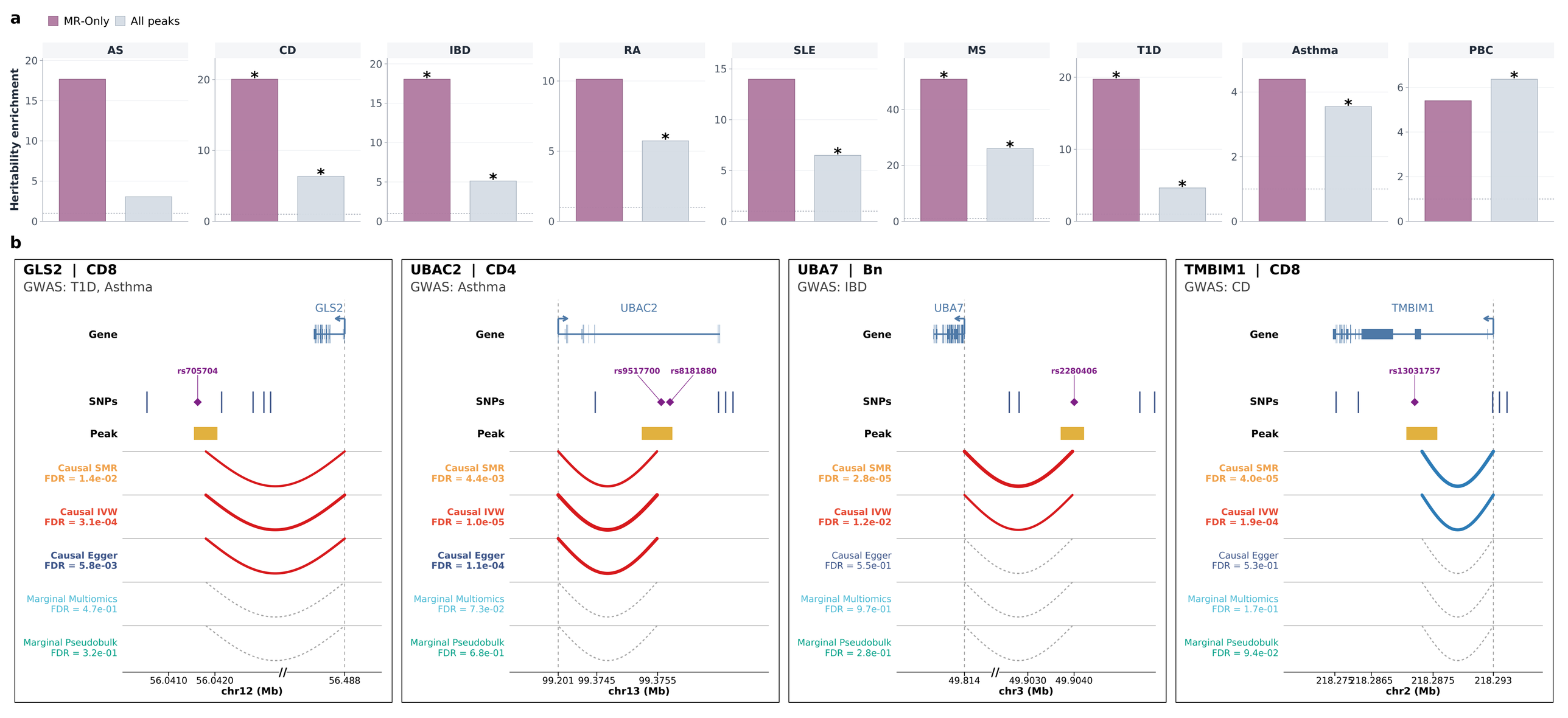
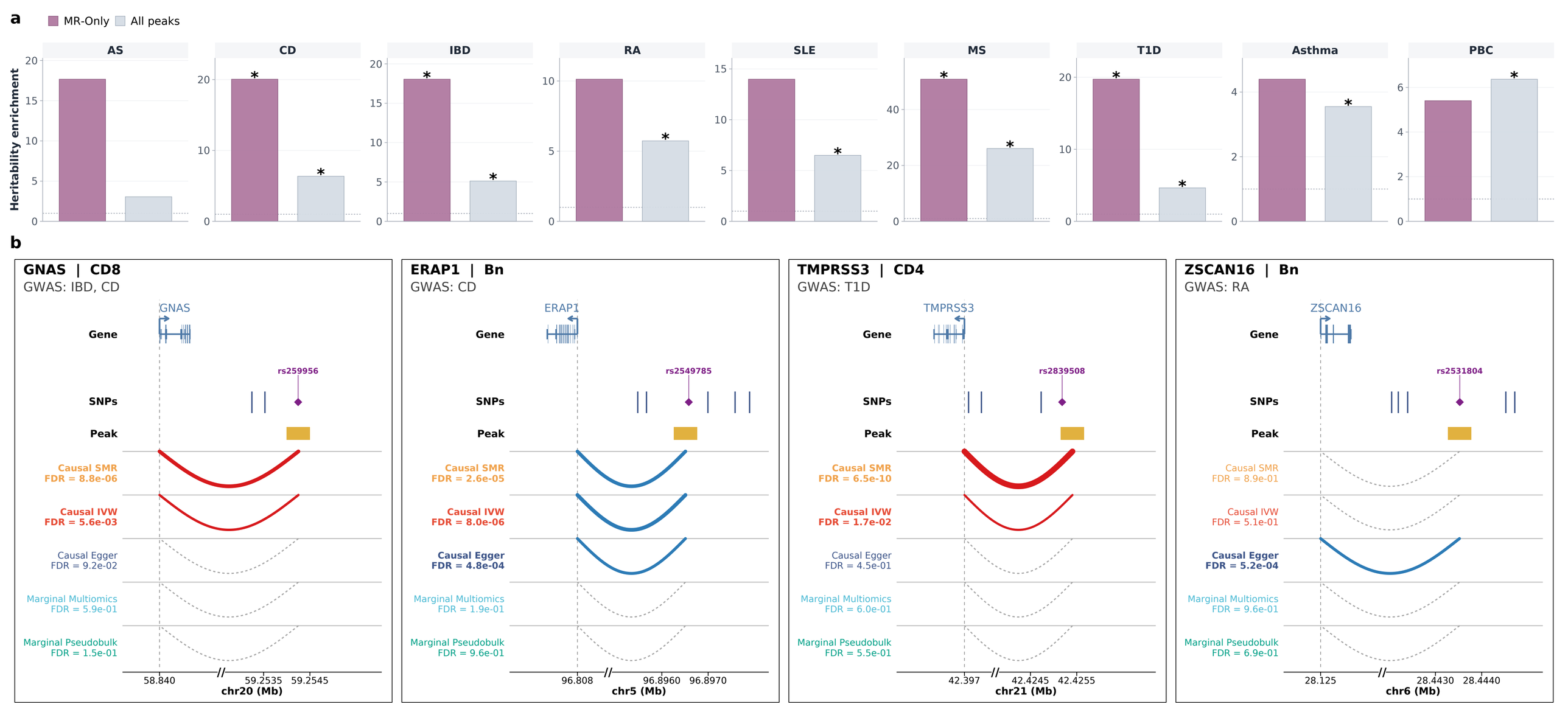

a
MR-Only
All peaks
Heritability enrichment
AS
CD
IBD
RA
SLE
MS
T1D
Asthma
PBC
b
RGL2 | Bn
GWAS: T1D, RA
RGL2
Gene
rs3106189
rs3130099
SNPs
Peak
Causal SMR
FDR = 9.7e-02
Causal IVW
FDR = 5.5e-04
Causal Egger
FDR = 1.3e-02
Marginal Multiomics
FDR = 5.2e-01
Marginal Pseudobulk
FDR = 8.2e-01
33.299
33.3135
33.3145
chr6 (Mb)
ERBB3 | CD4
GWAS: T1D, Asthma
ERBB3
Gene
rs705704
SNPs
Peak
Causal SMR
FDR = 2.9e-04
Causal IVW
FDR = 1.4e-04
Causal Egger
FDR = 1.3e-02
Marginal Multiomics
FDR = 4.4e-01
Marginal Pseudobulk
FDR = 1.2e-01
56.0410
56.0420
56.080
chr12 (Mb)
POLR2E | CD4
GWAS: CD
POLR2E
Gene
rs1046040
rs11084882
SNPs
Peak
Causal SMR
FDR = 8.1e-11
Causal IVW
FDR = 8.1e-04
Causal Egger
FDR = 1.1e-01
Marginal Multiomics
FDR = 9.4e-01
Marginal Pseudobulk
FDR = 5.7e-02
1.0945
1.0955
chr19 (Mb)
GNAS | CD4
GWAS: IBD, CD
GNAS
Gene
rs259956
SNPs
Peak
Causal SMR
FDR = 3.4e-04
Causal IVW
FDR = 7.3e-03
Causal Egger
FDR = 4.4e-02
Marginal Multiomics
FDR = 5.9e-01
Marginal Pseudobulk
FDR = 5.2e-01
58.840
59.2535
59.2545
chr20 (Mb)


a
MR-Only
All peaks
Heritability enrichment
AS
CD
IBD
RA
SLE
MS
T1D
Asthma
PBC
b
GLS2 | CD8
GWAS: T1D, Asthma
GLS2
Gene
rs705704
SNPs
Peak
Causal SMR
FDR = 1.4e-02
Causal IVW
FDR = 3.1e-04
Causal Egger
FDR = 5.8e-03
Marginal Multiomics
FDR = 4.7e-01
Marginal Pseudobulk
FDR = 3.2e-01
56.0410
56.0420
56.488
chr12 (Mb)
UBAC2 | CD4
GWAS: Asthma
UBAC2
Gene
rs9517700
rs8181880
SNPs
Peak
Causal SMR
FDR = 4.4e-03
Causal IVW
FDR = 1.0e-05
Causal Egger
FDR = 1.1e-04
Marginal Multiomics
FDR = 7.3e-02
Marginal Pseudobulk
FDR = 6.8e-01
99.201
99.3745
99.3755
chr13 (Mb)
UBA7 | Bn
GWAS: IBD
UBA7
Gene
rs2280406
SNPs
Peak
Causal SMR
FDR = 2.8e-05
Causal IVW
FDR = 1.2e-02
Causal Egger
FDR = 5.5e-01
Marginal Multiomics
FDR = 9.7e-01
Marginal Pseudobulk
FDR = 2.8e-01
49.814
49.9030
49.9040
chr3 (Mb)
TMBIM1 | CD8
GWAS: CD
TMBIM1
Gene
rs13031757
SNPs
Peak
Causal SMR
FDR = 4.0e-05
Causal IVW
FDR = 1.9e-04
Causal Egger
FDR = 5.3e-01
Marginal Multiomics
FDR = 1.7e-01
Marginal Pseudobulk
FDR = 9.4e-02
218.275
218.2865
218.2875
218.293
chr2 (Mb)


a
MR-Only
All peaks
Heritability enrichment
AS
CD
IBD
RA
SLE
MS
T1D
Asthma
PBC
b
GNAS | CD8
GWAS: IBD, CD
GNAS
Gene
rs259956
SNPs
Peak
Causal SMR
FDR = 8.8e-06
Causal IVW
FDR = 5.6e-03
Causal Egger
FDR = 9.2e-02
Marginal Multiomics
FDR = 5.9e-01
Marginal Pseudobulk
FDR = 1.5e-01
58.840
59.2535
59.2545
chr20 (Mb)
ERAP1 | Bn
GWAS: CD
ERAP1
Gene
rs2549785
SNPs
Peak
Causal SMR
FDR = 2.6e-05
Causal IVW
FDR = 8.0e-06
Causal Egger
FDR = 4.8e-04
Marginal Multiomics
FDR = 1.9e-01
Marginal Pseudobulk
FDR = 9.6e-01
96.808
96.8960
96.8970
chr5 (Mb)
TMPRSS3 | CD4
GWAS: T1D
TMPRSS3
Gene
rs2839508
SNPs
Peak
Causal SMR
FDR = 6.5e-10
Causal IVW
FDR = 1.7e-02
Causal Egger
FDR = 4.5e-01
Marginal Multiomics
FDR = 6.0e-01
Marginal Pseudobulk
FDR = 5.5e-01
42.397
42.4245
42.4255
chr21 (Mb)
ZSCAN16 | Bn
GWAS: RA
ZSCAN16
Gene
rs2531804
SNPs
Peak
Causal SMR
FDR = 8.9e-01
Causal IVW
FDR = 5.1e-01
Causal Egger
FDR = 5.2e-04
Marginal Multiomics
FDR = 9.6e-01
Marginal Pseudobulk
FDR = 6.9e-01
28.125
28.4430
28.4440
chr6 (Mb)